

Closed-loop resonant fiber optic gyro with an improved digital serrodyne modulation

Zhonghe Jin, Xuhui Yu, and Huilian Ma*

Micro-satellite Research Center, Zhejiang University, Hangzhou 310027, China
mahl@zju.edu.cn

Abstract: To widen the linear dynamic range and improve the linearity, a closed-loop resonant fiber optic gyro (RFOG) is proposed and experimentally demonstrated. To overcome the effect of the imperfect serrodyne modulation, an improved frequency shifting module is designed and constructed on a LiNbO₃ phase modulator. Its frequency resolution is improved to 0.01Hz which is equivalent to a rotation rate of 0.04°/h for an RFOG with a 12-cm diameter fiber ring resonator. With the frequency shifter applied in the RFOG, a closed-loop detection is demonstrated, whose bias stability is around 2 °/h, close to that of the open-loop output. Moreover, good linearity and wide dynamic range are also experimentally demonstrated thanks to the closed-loop operation. The measured result shows that the open-loop linear detection range of $\pm 215^\circ/\text{s}$ is improved to $\pm 1076^\circ/\text{s}$. It is improved by a factor of 5. The open-loop scale factor nonlinearity of 1.2% is decreased to 0.02% (200ppm), which is improved by a factor of 60. These are the best results reported to date, to the best of our knowledge, for closed-loop RFOGs.

©2013 Optical Society of America

OCIS codes: (120.5790) Sagnac effect; (060.2800) Gyroscopes; (230.5750) Resonators.

References and links

1. G. Sagnac, "L'ether lumineux demontre par l'effet du vent relatif d'ether dans un interferometre en rotation uniforme," C. R. Acad. Sci. **95**, 708–710 (1913).
2. H. Lefevre, *The Fiber-Optic Gyro* (Artech House, 1993).
3. K. Hotate, "Optical fiber sensors, applications, analysis, and future trends," in *Fiber-Optic Gyros*, J. Dakin and B. Culshaw, eds. (Artech House, 1997), pp. 167–206.
4. C. Caterina, P. Francesco, and M. N. Armenise, "A new integrated optical angular velocity sensor," Proc. SPIE **5728**, 93–100 (2005).
5. S. Ezekiel and R. Balsamo, "Passive ring resonator laser gyroscope," Appl. Phys. Lett. **30**(9), 478–480 (1977).
6. H. Ma, Z. He, and K. Hotate, "Reduction of backscattering induced noise by carrier suppression in waveguide-type optical ring resonator gyro," J. Lightwave Technol. **29**(1), 85–90 (2011).
7. H. Mao, H. Ma, and Z. Jin, "Polarization maintaining silica waveguide resonator optic gyro using double phase modulation technique," Opt. Express **19**(5), 4632–4643 (2011).
8. L. K. Strandjord and G. A. Sanders, "Performance improvements of a polarization-rotating resonator fiber optic gyroscope," Proc. SPIE **1795**, 94–104 (1993).
9. H. Ma, Z. Chen, Z. Yang, X. Yu, and Z. Jin, "Polarization-induced noise in resonator fiber optic gyro," Appl. Opt. **51**(28), 6708–6717 (2012).
10. X. Wang, Z. He, and K. Hotate, "Reduction of polarization-fluctuation induced drift in resonator fiber optic gyro by a resonator with twin 90° polarization-axis rotated splices," Opt. Express **18**(2), 1677–1683 (2010).
11. X. Wang, Z. He, and K. Hotate, "Automated suppression of polarization fluctuation in resonator fiber optic gyro with twin 90 polarization-axis rotated splices," J. Lightwave Technol. **31**(3), 366–374 (2013).
12. H. Ma, X. Yu, and Z. Jin, "Reduction of polarization-fluctuation induced drift in resonator fiber optic gyro by a resonator integrating in-line polarizers," Opt. Lett. **37**(16), 3342–3344 (2012).
13. X. Yu, H. Ma, and Z. Jin, "Improving thermal stability of a resonator fiber optic gyro employing a polarizing resonator," Opt. Express **21**(1), 358–369 (2013).
14. H. Ma, X. Lu, L. Yao, X. Yu, and Z. Jin, "Full investigation of the resonant frequency servo loop for resonator fiber-optic gyro," Appl. Opt. **51**(21), 5178–5185 (2012).
15. K. Hotate and G. Hayashi, "Resonator fiber optic gyro using digital serrodyne modulation-method to reduce the noise induced by the backscattering and closed-loop operation using digital signal processing," Proc. OFS-15, 104–107 (1999).
16. X. Wang, M. Kishi, Z. He, and K. Hotate, "Closed loop resonator fiber optic gyro with precisely controlled bipolar digital serrodyne modulation," Proc. SPIE **8351**, 83513G (2012).

17. L. K. Strandjord and G. A. Sanders, "Effects of imperfect serrodyne phase modulation in resonator fiber optic gyroscopes," *Proc. SPIE* **2292**, 272–282 (1994).
18. Z. Jin, X. Yu, and H. Ma, "Resonator fiber optic gyro employing a semiconductor laser," *Appl. Opt.* **51**(15), 2856–2864 (2012).
19. A. Ebbergh and G. Schiffner, "Closed-loop fiber-optic gyroscope with a sawtooth phase-modulated feedback," *Opt. Lett.* **10**(6), 300–302 (1985).
20. D. Ying, H. Ma, and Z. Jin, "Resonator fiber optic gyro using the triangle wave phase modulation technique," *Opt. Commun.* **281**(4), 580–586 (2008).
21. X. L. Zhang, H. I. Ma, Z. H. Jin, and C. Ding, "Open-loop operation experiments in a resonator fiber-optic gyro using the phase modulation spectroscopy technique," *Appl. Opt.* **45**(31), 7961–7965 (2006).

1. Introduction

It has been 100 years since the discovery of the Sagnac effect [1], which is the physical principle behind the operation of an optical gyro. One way of measuring the Sagnac effect is to use a so called Sagnac interferometer, which created the interferometric fiber optic gyro (IFOG) that has been developed for various applications. An IFOG employs a long fiber coil to enhance sensitivity. Their size and weight, as well as their cost, however, withhold them from applications requiring small, light and robust gyros [2–4]. Another way of measuring the Sagnac effect is by the use of an optical ring resonator, which created the resonant fiber optic gyro (RFOG) that was proposed and demonstrated early in 1977 by Ezekiel and Balsamo [5]. Lightwaves circulate many turns in the resonator, and thus enhance the Sagnac effect. Therefore, only a short length of sensing loop is needed for an RFOG. The inherent problem resulted from the use of a long fiber coil (~km) in the IFOG can be naturally overcome in the RFOG. However, compared with the IFOG, it is clear that more effort is still needed to make the RFOG practical.

The two major error sources in the RFOG have been identified as backscatter and polarization. These two types of noise limit the gyro sensitivity far greater than the shot noise associated with the photodetectors. The backscattering-induced noise is caused by the nonuniformity of the fiber which constitutes the ring resonator. It can be reduced below the shot noise limited level by the carrier-suppressed phase modulation technique [6, 7]. The polarization fluctuation-induced noise is dominantly caused by the existence of dual eigenstates of polarization (ESOPs) in the fiber resonator and by the temperature-sensitive birefringence of the fiber [8, 9]. Researchers have proposed several structures of the ring resonator with polarization maintaining fiber (PMF) and coupler [8–13]. An improved scheme for decreasing the polarization error by inserting two in-line polarizers in the PMF transmission-type resonator with twin 90° polarization-axis rotated splices has been proposed and demonstrated [12, 13]. A bias stability below 2°/h for an integration time of 100 s is successfully demonstrated in an RFOG with a PMF ring resonator having a ring length of 14.25 m.

The closed detection of an RFOG can improve the linearity and dynamic range of the gyro output. A high-sensitive IFOG always works in a closed manner. In the RFOG, the closed-loop operation includes two loops. The primary closed loop is used to cancel the fluctuations in the resonant frequency and the central frequency of the laser source. A detailed literature can refer to [14]. The secondary closed loop is designed for gyro rotation detection. A bipolar digital serrodyne phase modulation scheme has been developed on a single field-programmable gated array (FPGA) [15, 16]. The laser frequency is shifted through adding a serrodyne wave to the phase modulator. However, the resetting amplitude of the serrodyne wave is not ideal, which may cause an error in the RFOG [17]. In this paper, an improved frequency shifting module with a frequency resolution of 0.01 Hz, which is equivalent to a rotation rate of 0.04°/h, is proposed and demonstrated on a single FPGA. The effect of the imperfect serrodyne modulation has been overcome. At last, a gyro bias stability of 2 °/h is achieved in the closed-loop RFOG. Moreover, good linearity and large dynamic range are also experimentally demonstrated thanks to the closed-loop operation.

2. Principle and analysis

2.1 Closed-loop RFOG

Figure 1 shows a basic configuration of the closed-loop RFOG based on the sinusoidal phase modulation. The PMF transmission-type resonator with twin 90° polarization-axis rotated splices is the key rotation sensing element in the RFOG. For further decreasing the polarization error, two in-line polarizers P_x and P_y are inserted [12]. A lightwave from a narrow-linewidth fiber laser (linewidth less than 5 kHz) is divided into two equivalent beams by coupler C3. The LiNbO₃ phase modulators PM1 and PM2 are driven by sinusoidal waves with modulation frequencies f_1 and f_2 , respectively. The amplitudes of the two sinusoidal voltages V_1 and V_2 are carefully calibrated to suppress the carrier [6, 7]. The CW and the CCW lightwaves from the resonator are detected by the InGaAs PIN photodetectors, PD1 and PD2, respectively. The output of PD2 is fed back through the lock-in amplifier LIA2 to the servo controller PI1 to reduce the reciprocal noises in the RFOG [14, 18], which is the primary closed loop. To make the CW lightwave working in resonance, the demodulated signal of the CW lightwave from LIA1 is fed back to the LiNbO₃ phase modulator PM3 via the servo controller PI2 and the frequency shifting driver (FSD), which is the secondary closed loop for gyro signal detection. The frequency shifting function cannot be implemented by PM3 directly and the FSD module is inserted here to induce a frequency shifting signal. The frequency control word (FCW) is proportional to the shifted frequency or the frequency difference between the CW and CCW lightwaves, which is used as the readout of the rotation rate via a low-pass filter (LPF).

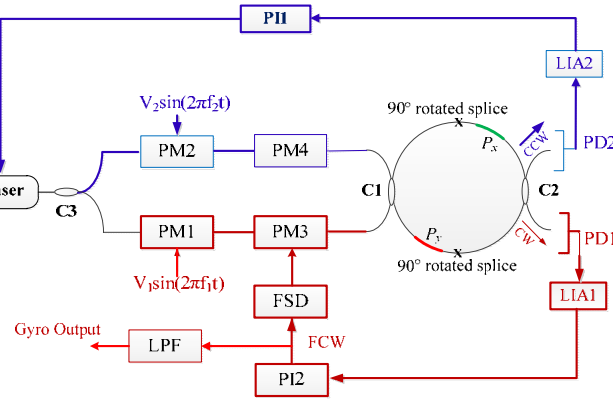


Fig. 1. Experimental setup of the closed-loop RFOG.

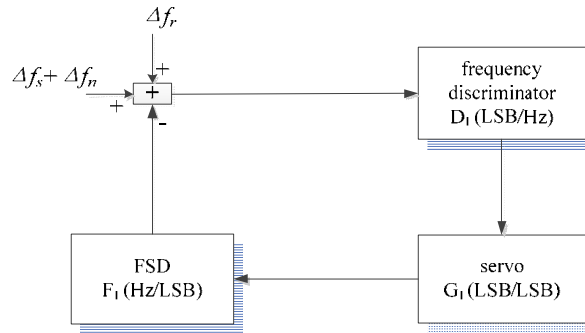


Fig. 2. Analysis model of the closed-loop RFOG.

Figure 2 shows the analysis model of the closed-loop RFOG. It consists of four main parts: input signal, frequency discriminator, servo, and the FSD model used to shift the laser frequency. The input signal includes three parts: the resonant frequency difference between the CW and the CCW lightwaves, Δf_s , contributing from the Sagnac effect; the nonreciprocal noise, Δf_n , contributing from the polarization fluctuation, Kerr effect and so on; the reciprocal noise, Δf_r , inducing by the residual laser frequency noise. The Sagnac effect and the nonreciprocal noise can't be distinguished for they are both generated in the optical fiber ring resonator (OFRR). As a result, Δf_s and Δf_n are added at the same point as shown in Fig. 2. The residual laser frequency noise has been suppressed effectively through the primary closed loop [18]. Therefore, the effect to the closed-loop detection caused by Δf_r can be ignored. The frequency discriminator, based on the phase modulation spectrum technology, converts the optical frequency fluctuations into the voltage fluctuations. The transfer function of the frequency discriminator is given by [18]

$$D_1 = k_{PD1} k_{LIA1}, \quad (1)$$

where the discriminative slope k_{PD1} from PD1 is amplified by the LIA1 gain k_{LIA1} .

The servo is the gain element that amplifies the voltage fluctuations provided by LIA1 and drives the FSD module. A simple proportional integral (PI) is applied, then the transfer function of the servo is given by [14]

$$G_1(s) = k_{PI} \left(1 + \frac{1}{\tau_{II}s}\right) \frac{1}{1 + \tau_{LPI}s}, \quad (2)$$

Where k_{PI} is the proportional gain and τ_{II} is the integration time. For convenience, the low-pass filter of LIA1 is also added here, and τ_{LPI} is its time constant. The FSD module is the signal generator that generates a serrodyne signal and drives PM3. Therefore, the frequency function can be implemented by PM3. The transformation factor $F1$ of the FSD model is equal to 0.01Hz/LSB, where LSB is the least significant bit. The calculation process of $F1$ will be given in the following section.

Using the aforementioned expressions, the open-loop transfer function for the model is written by

$$H_1(s) = D_1 G_1(s) F_1, \quad (3)$$

In the closed-loop operation, the transfer function of the output of the servo to the input is written by

$$T_1(s) = \frac{1}{F_1} \frac{H_1(s)}{1 + H_1(s)}, \quad (4)$$

Figure 3 shows the frequency response of the servo controller. As seen in Fig. 3, it shows a low-pass characteristic to the Sagnac effect and the nonreciprocal noise, and its response bandwidth is about 55 Hz. Therefore, the servo can detect the gyro output induced by the Sagnac effect, but can't suppress the nonreciprocal noise.

The transfer function of the output of the frequency discriminator to the input in the closed loop is written by

$$E_{D1}(s) = \frac{D_1}{1 + H_1(s)}, \quad (5)$$

It shows a high-pass characteristic to the input as shown in Fig. 4. The frequency noise suppression of 48 dB has been achieved at the frequency of 1 Hz, and it can suppress the low-frequency noise effectively. The frequency of the lightwave tracks the resonance of the OFRR in the closed loop through the servo. Thus, the output of the frequency discriminator tends to zero, and it also can be used to judge the performance of the loop.

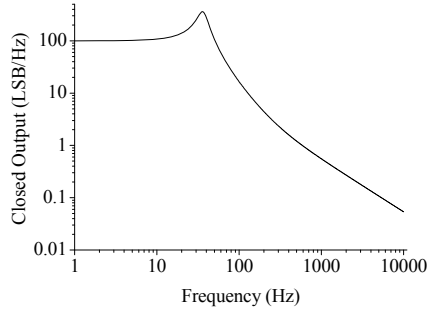


Fig. 3. Frequency response of the servo controller.

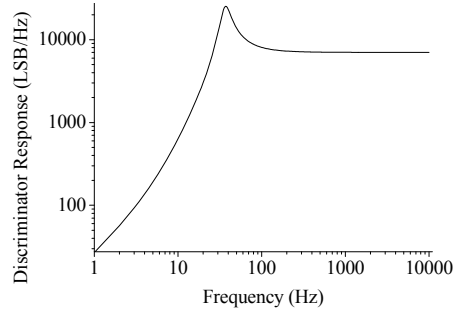


Fig. 4. Frequency response of the discriminator.

2.2 An improved digital serrodyne modulation applied with a phase modulator

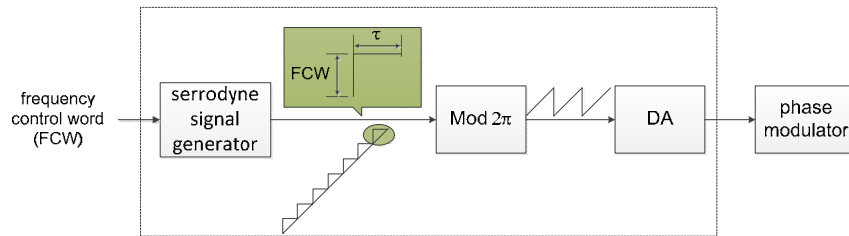


Fig. 5. Improved equivalent frequency shifting module.

A serrodyne waveform with amplitude of exact 2π can change the laser frequency ideally. The shifted frequency is equal to the slope of the serrodyne waveform [19]. The closed-loop operation has been achieved by superimposing an additional digital serrodyne waveform with a gentle slope to the original one [15, 16]. The serrodyne frequency shift is determined by the amplitude of the additional serrodyne waveform in a fixed period of time [15, 16]. Error appears when the varied amplitude of the combined serrodyne waveform deviates from the ideal value of 2π [17]. To overcome the effect of the imperfect serrodyne phase modulation and improve the frequency resolution, we design an improved frequency shifting module shown in Fig. 5. FCW is the stair height of the digital serrodyne waveform, which determines the shifted frequency. The amplitude of the serrodyne waveform is limited in the range of 2π by an additional mod operation module followed by the serrodyne signal generator, which is used to overcome the influence of the imperfect serrodyne phase modulation. At last, the digital serrodyne wave is converted to an analog one via a digital-to-analog (DA) converter, which drives the phase modulator.

The shifted frequency f_s , which is determined by the FCW, is given by

$$f_s = g \frac{FCW}{\tau}, \quad (6)$$

where τ is the stair duration of the digital serrodyne waveform and g is a conversion factor. Related to the reference voltage V_{ref} of the DA and the half-wave voltage V_π of PM3, g can be expressed as

$$g = \frac{V_{ref}}{2^B \times V_\pi}, \quad (7)$$

where B is the bits of the FCW. To improve the frequency resolution, a 32-bit FCW is used here and the frequency resolution of 0.01 Hz has been achieved according to Eq. (6), which is equivalent to a rotation rate of 0.04 °/h in the RFOG with a 12-cm diameter OFRR.

The amplitude of the serrodyne waveform plays an important role in the performance of the frequency shifter. When a serrodyne waveform, with a peak amplitude of $2\phi_0$ and with a near instantaneous fly-back time, is added to the phase modulator, the Fourier coefficients F_n of the carrier are given by [20]

$$F_n = \frac{\sin(\phi_0 - n\pi)}{\phi_0 - n\pi}, \quad (8)$$

where n is an integer, and it represents the harmonic components after being modulated. In the ideal case when ϕ_0 is exactly equal to π , all the optical power is shifted into a signal sideband where n is equal to 1. If ϕ_0 derives from π , a portion of optical power will be converted into spurious sidebands. The sideband suppression is given by

$$S_c = 10 \lg \left| \frac{F_1}{F_0} \right|^2, \quad (9)$$

when ϕ_0 is equal to π , the maximum sideband suppression can be achieved.

2.3 Measurements of the equivalent frequency shift

Figure 6 shows a schematic diagram of a Mach-Zehnder interferometer used for frequency shifting evaluation. A lightwave from a fiber laser is equally divided by the coupler C1. One arm of the interferometer contains an acoustic optic frequency shifter (AOFS), which provides a 40 MHz frequency shift to the lightwave. A phase modulator is placed within the second arm, which is driven by the FSD model. The two beams, after traversing their respective modulators, are recombined at the output coupler C2 and then launched into a photodetector (PD). This simple heterodyne detection allows us to measure the optical output spectrum by an electrical spectrum analyzer (ESA).

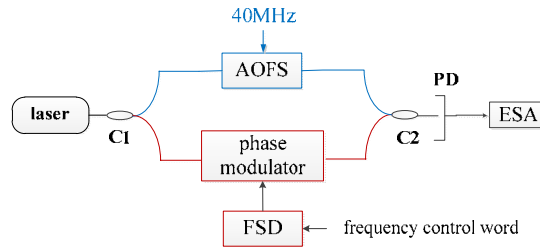


Fig. 6. Schematic diagram of a Mach-Zehnder interferometer used for frequency shifting evaluation.

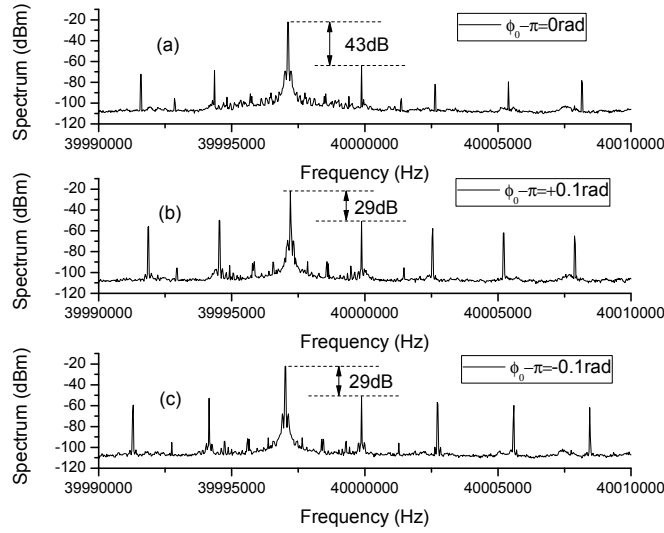


Fig. 7. Measurement results of the sidebands suppression with different amplitudes of the serrrodyne waveform. (a) $\phi_0 = \pi$. (b) $\phi_0 = \pi + 0.1\text{rad}$. (c) $\phi_0 = \pi - 0.1\text{rad}$.

When the FCW is set to 0, the optical power is shifted into the frequency of 40 MHz through the interferometer, which is observed by the ESA. The maximum sideband suppression has been achieved when the amplitude of the serrrodyne waveform is adjusted to close to π . In the ESA, the optical power is shifted by 2.75 kHz from the left of 40 MHz, when the FCW is set to 2^{18} , as shown in Fig. 7(a). In theory, the shifted frequency is 2.758 kHz according to Eq. (6), which is consistent with the experimental result. When the amplitude deviates π by $+0.1$ rad and -0.1 rad, both sideband suppressions of 29 dB have been achieved as shown in Figs. 7(b) and 7(c). According to Eq. (9), the sideband suppressions for the amplitude of $\pi + 0.1$ rad and $\pi - 0.1$ rad are 29.7 dB and 30.2 dB respectively, which are consistent with those experimental results.

Figure 8 shows the relationship between the shifted frequency and the FCW. The frequency shifting range is ± 1.25 MHz and the slope of this linear fitting curve is 0.01Hz/LSB, which is equal to the frequency resolution of the frequency shifter.

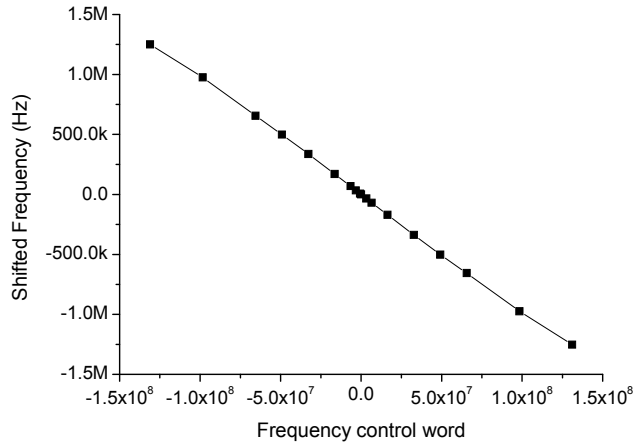


Fig. 8. Relationship between the serrrodyne shifted frequency and the FCW.

To test the frequency resolution of the frequency shifter, another PM instead of the AOFs, is placed within the first arm of the interferometer in Fig. 6. Therefore, a shifted frequency of

0.01 Hz can be detected at a low-intermediate frequency. A serrrodyne waveform, generated by a signal generator, with a peak amplitude of 2π and a frequency of 1 kHz, drives the PM in the first arm. Figure 9 shows the measurement optical spectrum at three different FCW values, namely 0, 1 and 2. When the FCW is set to 0, the optical spectrum should be theoretically centered at 1 kHz. However, the observed spectrum on the ESA as shown in Fig. 9(a) is centered at 1000.0100 Hz. When the FCW is set to 1, the observed spectrum on the ESA is shifted 0.0075 Hz from the right of 1000.0100 Hz, as shown in Fig. 9(b). In theory, the shifted frequency is 0.01 Hz according to Eq. (6). The aforementioned difference between the experimental results and those theoretical ones is considered as the effect of the limited resolution of the ESA. When the FCW is set to 2, the measured shifted frequency is 0.02 Hz shown in Fig. 9(c), which is consistent with the theoretical one. Therefore, the frequency resolution of 0.01 Hz is verified, which is equivalent to a rotation rate of $0.04^\circ/\text{h}$ with a 12-cm OFRR.

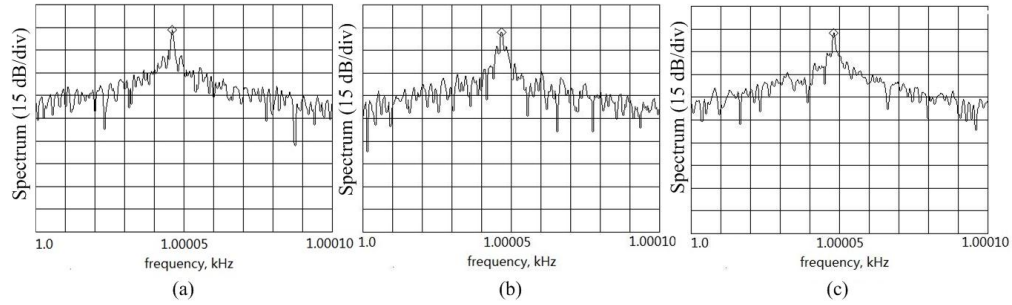


Fig. 9. Measurements of optical spectrum using the MZI. (a) Spectral display at 1000.010 Hz with the FCW of 0. (b) Spectral display at 1000.0175 Hz with the FCW of 1. (c) Spectral display at 1000.030 Hz with the FCW of 2.

3. Measurements of the closed-loop RFOG

3.1 Bias stability

The experimental setup of the closed-loop RFOG is shown in Fig. 1. To make the frequency of the lightwave to track the resonance of the OFRR in the CW direction quickly, the proper parameters of PI2 are adopted here. When the secondary closed loop works, the average output from LIA1 is close to zero for the integration time of 1 s, as shown in Fig. 10(a). As a result, the CW lightwave is in resonance. The Allan deviation of the output from LIA1 is shown in Fig. 10(b). The calculated residual frequency noise is equivalent to a rotation rate of $0.2^\circ/\text{h}$ for the integration time of 10 s, and close to the shot noise limit for the RFOG. The Allan deviation is dominated by the white frequency noise ($\propto \tau^{-1/2}$). Thus, the closed loop works well.

The FCW is proportional to the shifted frequency or the frequency difference between the CW and the CCW lightwaves, which is used as the readout of the rotation rate via an LPF. Figure 11(a) shows the closed-loop output tested from the LPF for the integration time of 1 s. The Allan deviation of the closed-loop output is shown in Fig. 11(b). The gyro bias stability is about $2^\circ/\text{h}$, approximately equal to that of the open-loop output [13].

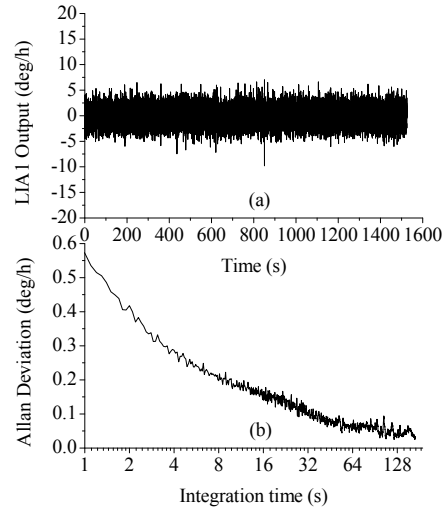


Fig. 10. Output from LIA1 in the secondary closed loop.

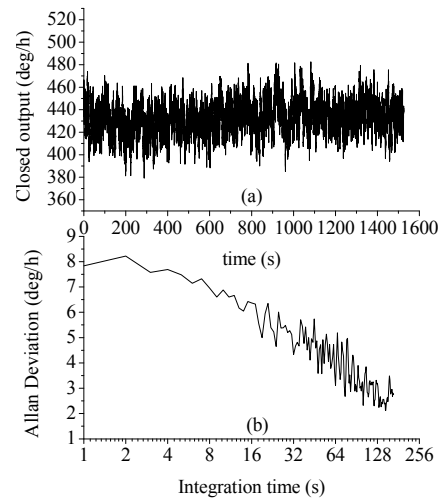


Fig. 11. Measurement results of the closed-loop RFOG.

The swing rotation measurement results of the closed-loop RFOG are shown in Fig. 12. The rotation amplitudes of the sinusoidal swing are $\pm 0.005^\circ/\text{s}$, $\pm 0.05^\circ/\text{s}$, $\pm 0.5^\circ/\text{s}$ and $\pm 5^\circ/\text{s}$, respectively. Obviously, the swinging rate of $\pm 0.005^\circ/\text{s}$ ($18^\circ/\text{h}$) can be observed easily, which is close to the earth rotation rate of $15^\circ/\text{h}$.

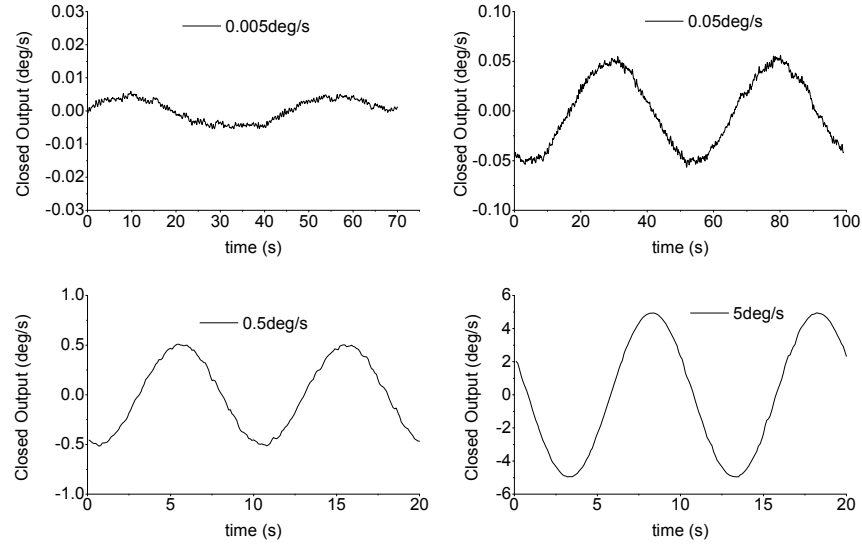


Fig. 12. Swing rotation response of the closed-loop RFOG.

3.2 Dynamic range and linearity

To test the linearity and dynamic range of the RFOG, two acoustic optic frequency shifters, AOFS1 and AOFS2, are added after phase modulators in Fig. 1. A sinusoidal signal, with the fixed frequency of 55 MHz, drives AOFS1. At the same time, another sinusoidal signal, with the frequency adjusted in the vicinity of 55 MHz, drives AOFS2. The frequency difference between the two sinusoidal signals is equal to the frequency difference between the CW and the CCW lightwaves. Therefore, a quasi-rotation is induced in the RFOG.

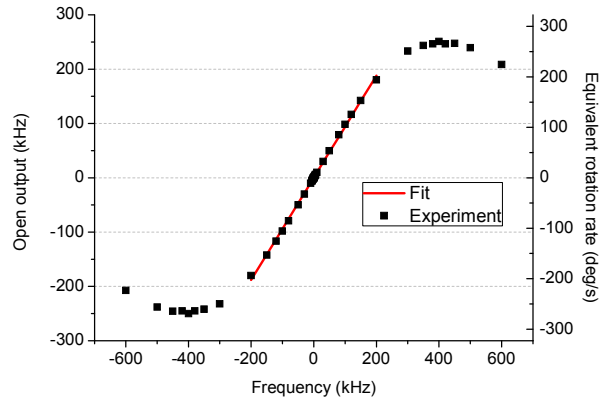


Fig. 13. Quasi-rotation measurement results of the open-loop RFOG.

The demodulated signal of the CW lightwave from LIA1 is used as the open-loop output of the rotation rate if it is not used to fed back to PM3. Figure 13 shows the quasi-rotation measurement results of the open-loop RFOG. The demodulated output increases as the rotation rate increases in a range of ± 400 kHz. For the quasi-rotation of 400 kHz, a demodulated output of 251 kHz has been achieved. The linearity has become very bad at the frequency difference of 400 kHz. Therefore, the open-loop linear detection range of ± 200 kHz is achieved, which is equivalent to a rotation range of $\pm 215^\circ/\text{s}$. After linear fitting, the

nonlinearity of the scale factor is about 1.2% in the range of ± 200 kHz, as indicated in red line in Fig. 13. The linear detection range and scale factor nonlinearity for the open-loop detection are limited by the phase modulation spectrum technology [21].

When the secondary closed loop works, the closed gyro output is proportional to the frequency difference between the CW and the CCW lightwaves in a range of ± 1 MHz as shown in Fig. 14. Compared to the open-loop detection, the closed-loop RFOG has a wider linear detection range and lower scale factor nonlinearity. The measured result shows that the open-loop linear detection range of $215^\circ/\text{s}$ is improved to $\pm 1076^\circ/\text{s}$. It is improved by a factor of 5. The open-loop scale factor nonlinearity of 1.2% is decreased to 0.02% (200ppm), which is improved by a factor of 60. The excellent linearity and large dynamic range of the gyro output is beneficial from the closed-loop operation, because the gyro signal is read as a frequency signal directly from PI2, not as a voltage signal from LIA1.

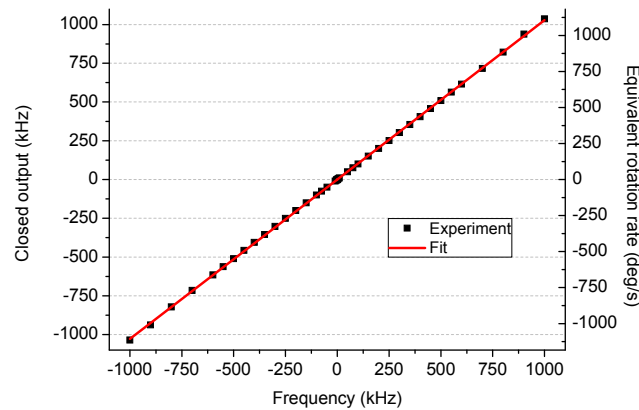


Fig. 14. Quasi-rotation measurement results of the closed-loop RFOG.

4. Conclusions

A closed-loop RFOG is demonstrated in this paper. We design an improved frequency shifting module with a high frequency resolution of 0.01Hz. The effect of the imperfect serrodyne modulation can be overcome. The gyro bias stability of $2^\circ/\text{h}$ is achieved and the scale factor nonlinearity decreased to 200ppm in a range of $\pm 1076^\circ/\text{s}$ is also made. The excellent linear and large dynamic range of the gyro output is beneficial from the closed-loop operation.

Acknowledgment

The authors would like to acknowledge financial support from the National Natural Science Foundation of China (No. 61377101).